



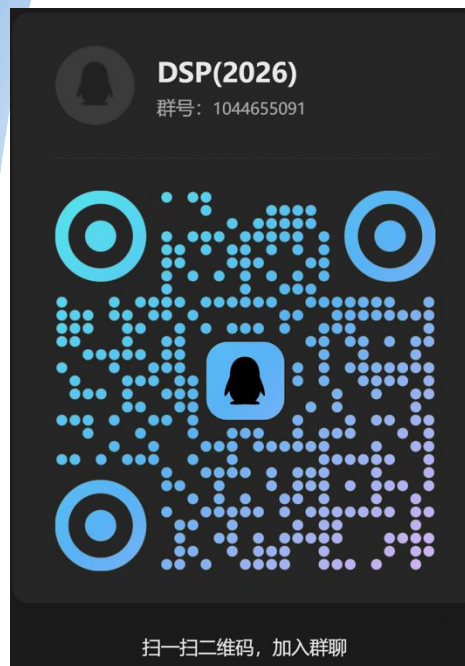
Digital Signal Processing

Lecturer : 利强

Office: 科研楼C211

Email: LQ@uestc.edu.cn

TA: 杨柳





MOOC Website:

<https://www.icourse163.org/course/UESTC-236010>

<https://www.icourse163.org/course/UESTC1-1458252163>

中国大学MOOC

课程

学校

数字教材

慕课堂

升学考研

下载APP

26考研全科规划



首页 > Electronics

Digital Signal Processing

播放

Galasgow college&Infor. & Comm. Eng. college, UESTC

Digital Signal Processing

第7次开课

开课时间: 2025年02月25日 ~ 2025年06月30日

学时安排: 3-4小时每周

距离开课还有 7 天

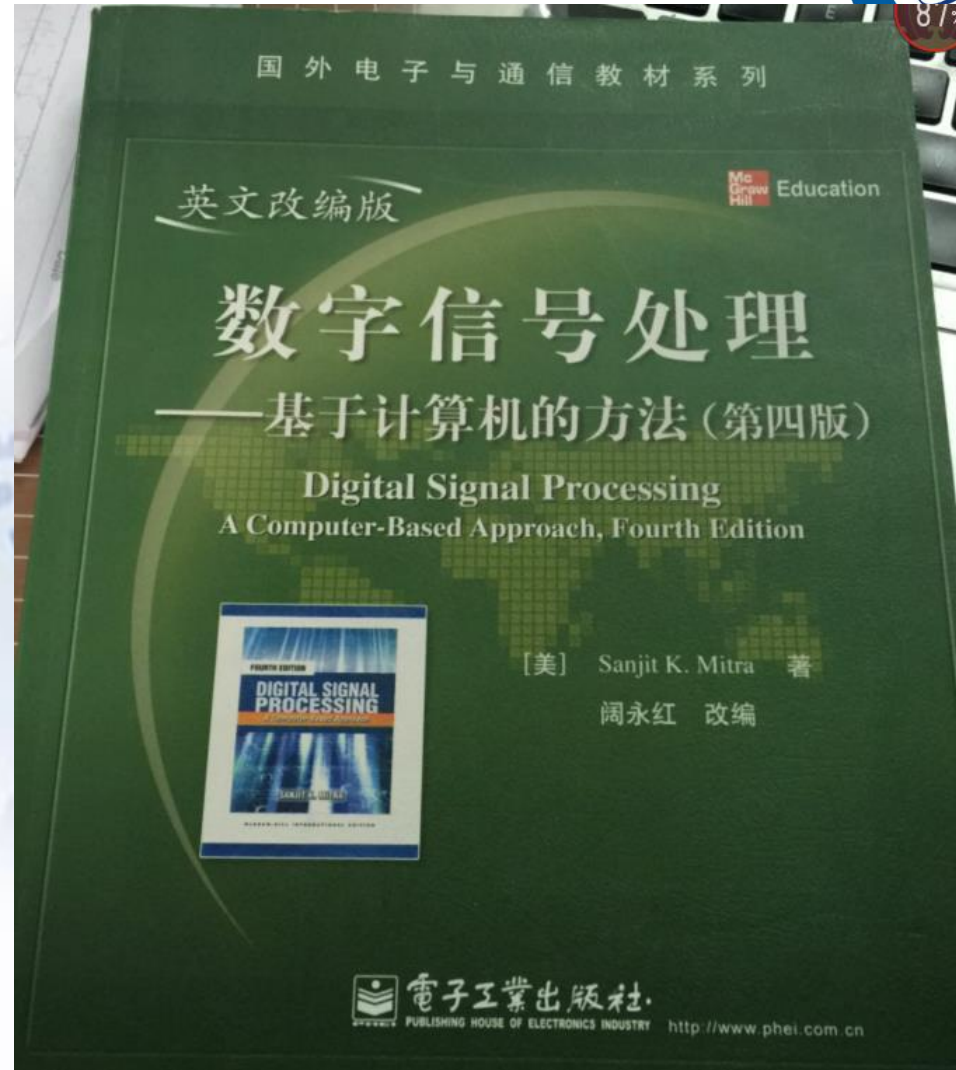
已参加, 等待开课

The Textbook



**Digital Signal
Processing---A
Computer- Based
Approach, Sanjit
K.Mitra,4th
Edition,McGraw-
Hill,2014**

电子工业出版社



Reference books



- 1. 奥本海姆（李玉柏等译），离散时间信号处理（原书第3版.精编版），机械工业出版社
- 2. 彭启琮等，数字信号处理, 高等教育出版社
- 3. 彭启琮等,DSP与实时数字信号处理,电子科大出版社
- 4. 程佩青,数字信号处理教程,清华大学出版社
- 5.胡广书，数字信号处理——理论、算法与实现，清华大学出版社

Arrangement



1. **Lectures**, 56 chs. (including course contents 48 chs. and presentations of course projects 8 chs.)
2. **MOOC**, about 32 chs.
3. **Lab time**, about 8 chs.

Grading



1. Course projects:25%

(1) Complete the design as required (Include MATLAB codes, graphs, result analysis etc.)

(2) Presentation

(3) Project report

2. Open Book Final Examination: 75%

Only one exam

Total 6 questions in 3 hours

Refer to the questions in MOOC and Homework

Contents of Mooc



Mooc videos	Book chapters
Week1: DSP-introduction and MATLAB 2videos	Ch1
Week1: Digital Processing of Continuous Signals-2	Ch2
Digital Processing of Continuous Signals-1&3	Ch3.8
Digital Processing of Continuous Signals-4	Ch12
Week2-3: DTFT 5videos	Ch3.1-3.6
Week4: Discrete Time System 4videos	Ch4
Week5-6: DFT 4videos&FFT 2video	Ch5&Ch11
Week7: ZT and IZT 5videos	Ch6
Week8-9:LTI Discrete-Time System in the Transform Domain 5videos	Ch7
Week10: Digital Filter Structures 2videos	Ch8
Week11-12: FDA tools&filter design 7videos	Ch9-10
Final Examination begins from Week 13.	

Foreword

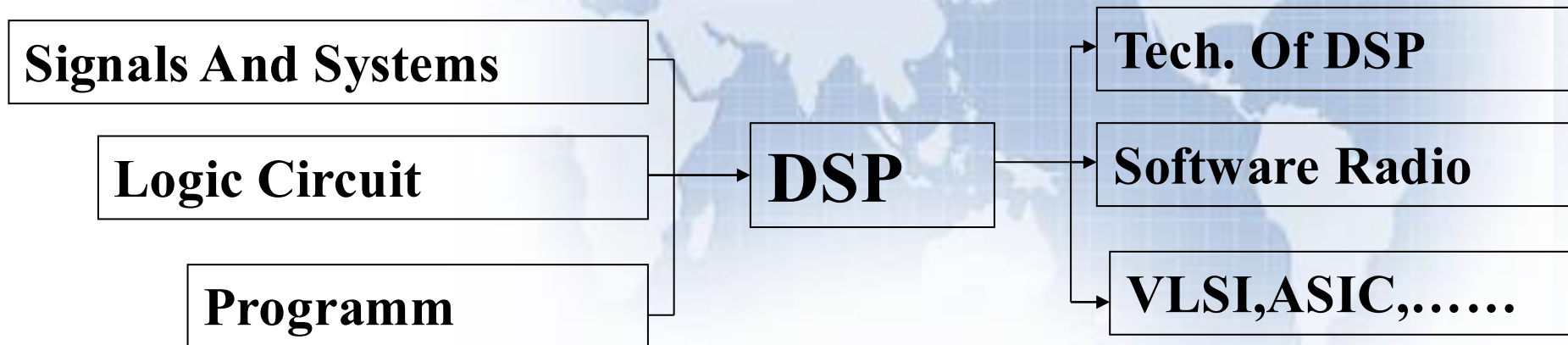


Characterization Of The Course

1.The Tech. Fields Used DSP:

Communication, Radio, Radar, Navigation, IT, etc

2. The Correlated Course:



Required Contents



- § 1
- § 3* 3.1~3.9
- § 5* 5.1~ 5.10
- § 7 7.1~7.4
- § 9 * 9.1~9.6
- § 11 11.1~11.5
- § 2 2.1~2.5
- § 4 4.1~ 4.10
- § 6 6.1~6.7
- § 8 8.1~8.6
- § 10 * 10.1,10.2,10.5
- § A A.1~A.8

1 Signals and Signal Processing	1
1.1 Characterization and Classification of Signals	1
1.2 Typical Signal Processing Operations	4
1.3 Examples of Typical Signals	13
1.4 Typical Signal Processing Applications	21
1.5 Why Digital Signal Processing?	29
2 Discrete-Time Signals in the Time Domain	33
2.1 Time-Domain Representation	34
2.2 Operations on Sequences	38
2.3 Operations on Finite-Length Sequences	47
2.4 Typical Sequences and Sequence Representation	54
2.5 The Sampling Process	64
2.6 Correlation of Signals	66
2.7 Random Signals	72
2.8 Summary	73
2.9 Problems	73
2.10 MATLAB Exercises	79
3 Discrete-Time Signals in the Frequency Domain	81
3.1 The Continuous-Time Fourier Transform	81
3.2 The Discrete-Time Fourier Transform	86
3.3 Discrete-Time Fourier Transform Theorems	97
3.4 Energy Density Spectrum of a Discrete-Time Sequence	103
3.5 Band-Limited Discrete-Time Signals	104
3.6 DTFT Computation Using MATLAB	105
3.7 The Unwrapped Phase Function	105
3.8 Digital Processing of Continuous-Time Signals	107

3.9 Sampling of Bandpass Signals	121
3.10 Effect of Sample-and-Hold Operation	123
3.11 Summary	124
3.12 Problems	125
3.13 MATLAB Exercises	134
4 Discrete-Time Systems	135
4.1 Discrete-Time System Examples	135
4.2 Classification of Discrete-Time Systems	141
4.3 Impulse and Step Responses	145
4.4 Time-Domain Characterization of LTI Discrete-Time Systems	146
4.5 Simple Interconnection Schemes	153
4.6 Finite-Dimensional LTI Discrete-Time Systems	156
4.7 Classification of LTI Discrete-Time Systems	164
4.8 Frequency-Domain Representations of LTI Discrete-Time Systems	167
4.9 Phase and Group Delays	174
4.10 Summary	178
4.11 Problems	179
4.12 MATLAB Exercises	187
5 Finite-Length Discrete Transforms	188
5.1 Orthogonal Transforms	188
5.2 The Discrete Fourier Transform	190
5.3 Relation Between the DTFT and the DFT and Their Inverses	194
5.4 Circular Convolution	200
5.5 Classifications of Finite-Length Sequences	205
5.6 DFT Symmetry Relations	210
5.7 Discrete Fourier Transform Theorems	213
5.8 Fourier-Domain Filtering	219
5.9 Computation of the DFT of Real Sequences	221
5.10 Linear Convolution Using the DFT	223
5.11 Summary	234
5.12 Problems	234

5.13 MATLAB Exercises	247
6 z-Transform	249
6.1 Definition	249
6.2 Rational z-Transforms	253
6.3 Region of Convergence of a Rational z-Transform	255
6.4 The Inverse z-Transform	261
6.5 z-Transform Theorems	269
6.6 Computation of the Convolution Sum of Finite-Length Sequences	277
6.7 The Transfer Function	280
6.8 Summary	292
6.9 Problems	292
6.10 MATLAB Exercises	304
7 LTI Discrete-Time Systems in the Transform Domain	305
7.1 Transfer Function Classification Based on Magnitude Characteristics	305
7.2 Transfer Function Classification Based on Phase Characteristics	314
7.3 Types of Linear-Phase FIR Transfer Functions	321
7.4 Simple Digital Filters	332
7.5 Inverse Systems	349
7.6 Summary	353
7.7 Problems	354
7.8 MATLAB Exercises	368
8 Digital Filter Structures	370
8.1 Block Diagram Representation	371
8.2 Equivalent Structures	374
8.3 Basic FIR Digital Filter Structures	375
8.4 Basic IIR Digital Filter Structures	380
8.5 Realization of Basic Structures Using MATLAB	386
8.6 Allpass Filters	389
8.7 IIR Tapped Cascaded Lattice Structures	398
8.8 FIR Cascaded Lattice Structures	403
8.9 Summary	408
8.10 Problems	409
8.11 MATLAB Exercises	422

9 IIR Digital Filter Design	424
9.1 Preliminary Considerations	424
9.2 Bilinear Transformation Method of IIR Filter Design	429
9.3 Design of Lowpass IIR Digital Filters	434
9.4 Design of Highpass, Bandpass, and Bandstop IIR Digital Filters	436
9.5 Spectral Transformations of IIR Filters	440
9.6 IIR Digital Filter Design Using MATLAB	447
9.7 Summary	450
9.8 Problems	450
9.9 MATLAB Exercises	456
10 FIR Digital Filter Design	458
10.1 Preliminary Considerations	458
10.2 FIR Filter Design Based on Windowed Fourier Series	462
10.3 Computer-Aided Design of Equiripple Linear-Phase FIR Filters	477
10.4 Design of Minimum-Phase FIR Filters	486
10.5 FIR Digital Filter Design Using MATLAB	487
10.6 Summary	503
10.7 Problems	504
10.8 MATLAB Exercises	511
11 DSP Algorithm Implementation	515
11.1 Basic Issues	515
11.2 Structure Simulation and Verification Using MATLAB	526
11.3 Computation of the Discrete Fourier Transform	533
11.4 Fast DFT Algorithms Based on Index Mapping	548
11.5 DFT and IDFT Computation Using MATLAB	556
11.6 Sliding Discrete Fourier Transform	558
11.7 DFT Computation over a Narrow Frequency Band	558
11.8 Summary	563
11.9 Problems	563
11.10 MATLAB Exercises	571
Analog Lowpass Filter Design	574
A.1 Analog Filter Specifications	574
A.2 Butterworth Approximation	576

Projects:



- **Production and Detection of Dual-Tone Multi-Frequency (DTMF) Signals**

In week 8

- **Digital Frequency Spectrum Analysis Based on DFT**

In week 10